



Risk-Averse Humanitarian Logistics

Interim Presentation | MGTHN0197 | Topic 5

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Today's plan

1. Motivation and research question
2. What we build (model sketch)
3. Case study: Rawls Gulf Coast
4. Workflow from data to results
5. Preliminary results (RN vs RA)
6. Plan for the final presentation

1. Motivation and research question

- ▶ After a disaster, people need water, food and medicine within hours.
- ▶ Hub locations and stock levels are chosen *before* the storm.
- ▶ A plan that looks good on average can fail in the worst hurricanes.

Research question

How does moving from a risk-neutral plan to a CVaR-based risk-averse plan change unmet demand and early response?

2. Model sketch

Two-stage stochastic MILP

- ▶ Stage 1: open hubs and pre-position stock.
- ▶ Stage 2: ship by truck ($\tau = 1$) and air ($\tau = 0$), track inventory and unmet demand.

Objective (mean-CVaR on unmet demand)

$$\min \sum_s p_s Z_s + \lambda \left[\eta + \frac{1}{1-\alpha} \sum_s p_s \xi_s \right]$$

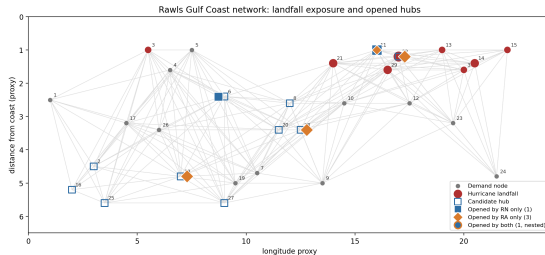
with $\xi_s \geq S_s - \eta$, S_s = total unmet demand in scenario s .

Scope

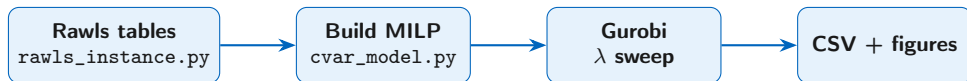
Mean-CVaR stochastic programming. Distributionally robust models are out of seminar scope.

3. Case study: Rawls Gulf Coast

- ▶ 30 nodes, 11 candidate hubs.
- ▶ 15 single-hurricane scenarios from Rawls and Turnquist (2010).
- ▶ Damage rule: $\text{Cat} \geq 3$ destroys landfall stock, $\text{Cat} \leq 2$ leaves 50%.
- ▶ Unusable roads zero truck capacity. Air stays open.
- ▶ Commodities: water, food, medicine. Three periods (50/30/20).



4. Workflow: data $\rightarrow$ code $\rightarrow$ results



Policies

- ▶ Risk-neutral $\lambda = 0$.
- ▶ Risk-averse $\lambda = 18$, $\alpha = 0.95$.

Metrics

- ▶ $\mathbb{E}[Z]$, CVaR_{0.95} of unmet demand.
- ▶ Coverage by scenario, period, commodity.

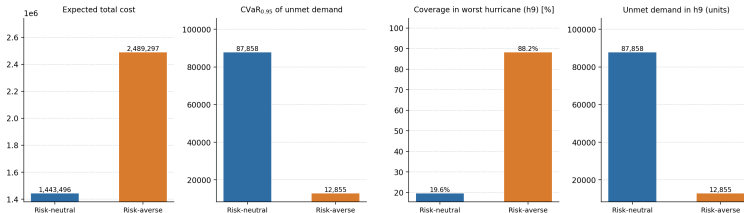
5. Preliminary results

Metric	RN	RA
$\mathbb{E}[Z]$	1,443,496	2,489,297
CVaR _{0.95} unmet	87,858	12,855
Expected coverage	83.5%	93.4%
Coverage in h9	19.6%	88.2%
Period-1 coverage in h9	0.9%	76.5%
Hubs opened	2	4
Solve time (s)	10.8	13.4

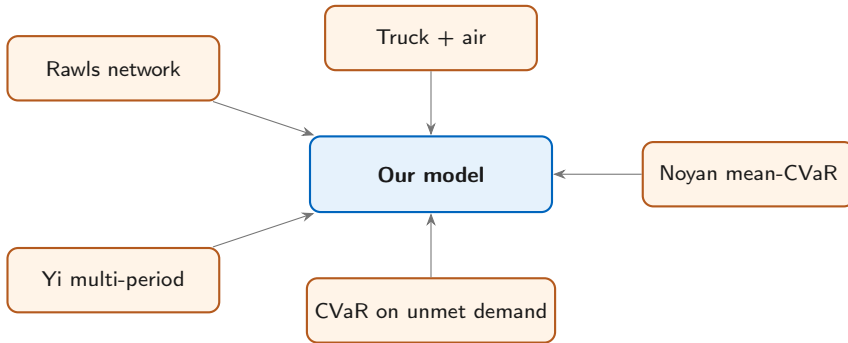
Takeaway

Cost +72%. Tail unmet demand –85%.
Worst-hurricane coverage
19.6% → 88.2%.

Policy comparison ($\lambda = 0$ vs $\lambda = 18$) — Rawls Gulf Coast



6. Contribution and next steps



For the final presentation

Full MILP (sets, parameters, variables, objective, constraints), CDF of unmet demand, λ -Pareto, managerial insights, and outlook.

References

- [1] Rawls, C. G., and Turnquist, M. A. (2010). Pre-positioning of emergency supplies for disaster response. *Transportation Research Part B: Methodological*, 44(4), 521–534.
- [2] Yi, W., and Özdamar, L. (2007). A dynamic logistics coordination model for evacuation and support in disaster response activities. *European Journal of Operational Research*, 179(3), 1177–1193.
- [3] Noyan, N. (2012). Risk-averse two-stage stochastic programming with an application to disaster management. *Computers & Operations Research*, 39(3), 541–559.

Thank you

Questions welcome. Repository.

<https://github.com/PNeDIT/mgthn0197-humanitarian-cvar>